

SAFOD DAS Session Handoff

What changed after the preflight output, what we learned, and what to run next

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Executive read

The first one-day run was a useful diagnostic, but not a final verdict on the 2024 data. It used daytime-only files, which is not stated in Lellouch et al. 2019. The next controlled tests are all-hours stacking, multi-day stacking, source-channel sweeps, and a same-code test on the 2017 data.

Current State In One Sentence

We have a cleaner SAFOD sanity pipeline, the one-day daytime CC did not reproduce a clear Fig. 7c-style body-wave moveout, and the immediate next work is to test whether that changes when using all continuous files, more days, different virtual-source channels, and finally the 2017 data.

Important Correction

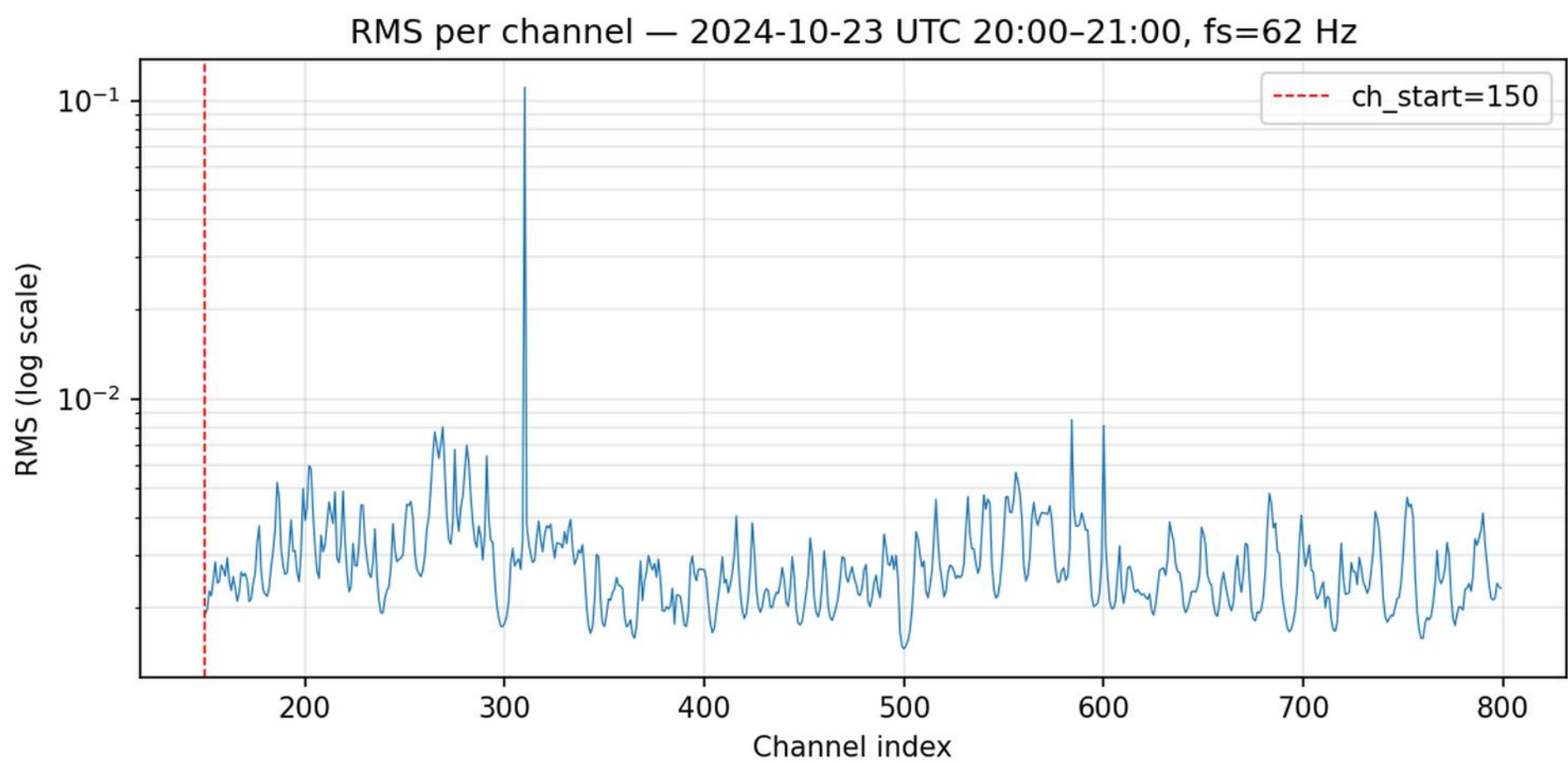
- Gauge length is not the likely blocker in the 5-20 Hz band. The finite-gauge sinc term is close to flat for both 3200 m/s body waves and ~1500 m/s guided/tube waves.
- The daytime-only file selection was our hypothesis about cultural noise, not something Lellouch explicitly described.
- The vertical fiber geometry makes standard lateral surface-wave hydrology dv/v a poor fit, but does not rule out body-wave, coda, repeating-earthquake, or carefully interpreted guided-wave monitoring.

Timeline Since Preflight

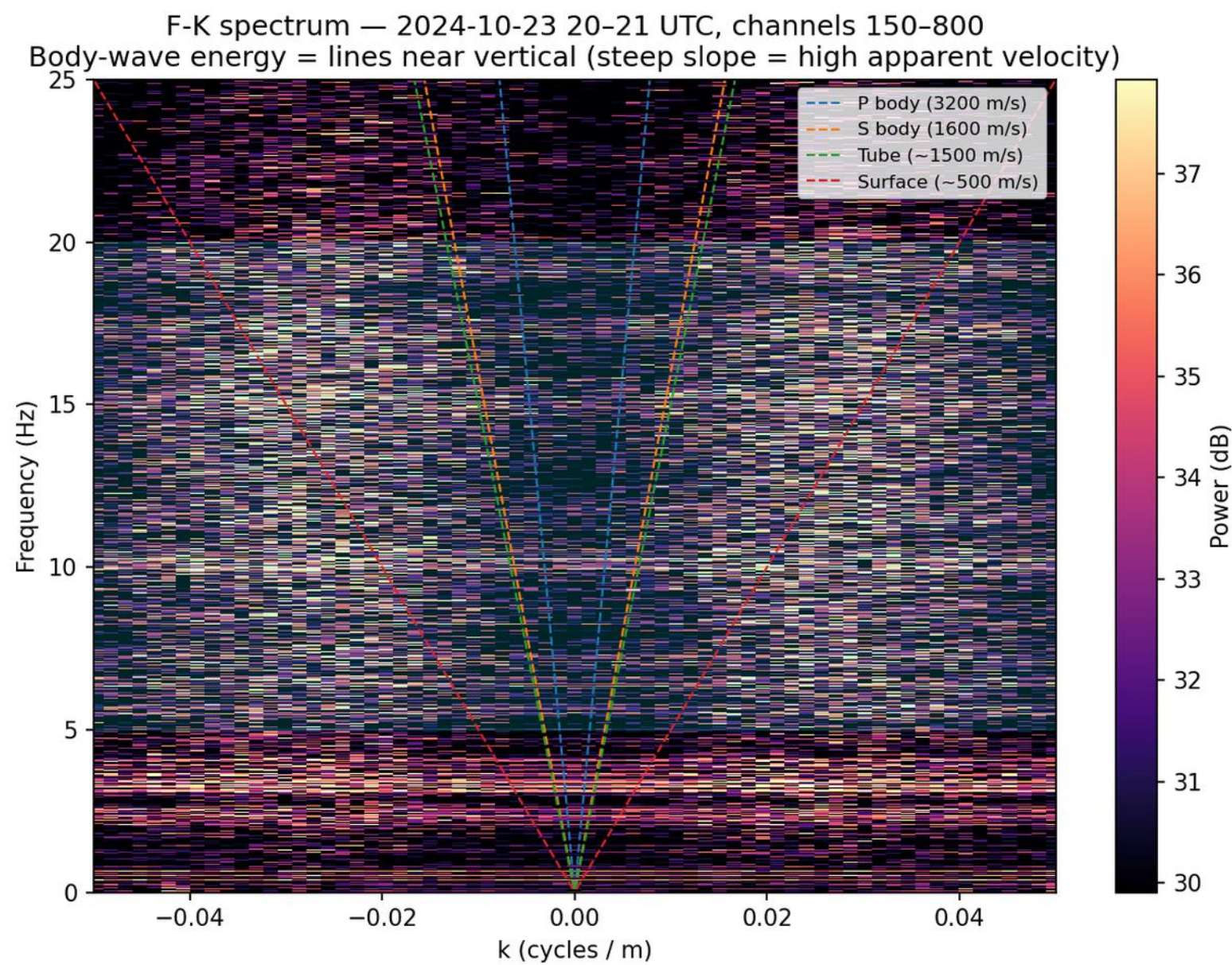
Step	What Happened	Interpretation
Preflight outputs inspected	Looked at RMS, time-distance, and F-K plots for 2024-10-23 20-21 UTC, channels 150-800.	Preflight justified continuing to CC, but it was not by itself proof of a recovered body-wave Green's function.
One-day CC completed	SLURM job 24589057 ran the CC stage for 2024-10-23 with ch 150-800, source ch 150, 5-20 Hz, 30 s windows, 50% overlap, running-AM 0.1 s, whitening off.	The run selected 600 daytime continuous files and produced n_stack=1800. Important caveat: daytime-only was our assumption, not stated in Lellouch.
One-day plots reviewed	Raw and R+/-10 pre-shifted stacked plots were dominated by near-zero-lag vertical energy, with no clean arrival tracking the 3200 m/s line.	This did not reproduce Lellouch Fig. 7c, but it is only a daytime one-day result, not a final statement about 2024.
Follow-up scope corrected	Decided not to pivot after one failed run. Added tests for more days, alternate virtual-source channels, and same-code testing on 2017 data.	This is the right scientific escalation: rule out implementation and selection effects before interpreting 2024 as different.
Hour selection corrected	Recognized that Lellouch described one day of continuous data, not daytime-only. Added a local code change for SANITY_HOURS=all day night, default all.	The strict reproduction path should use all continuous files. Day/night become explicit source-distribution tests.

Preflight Evidence

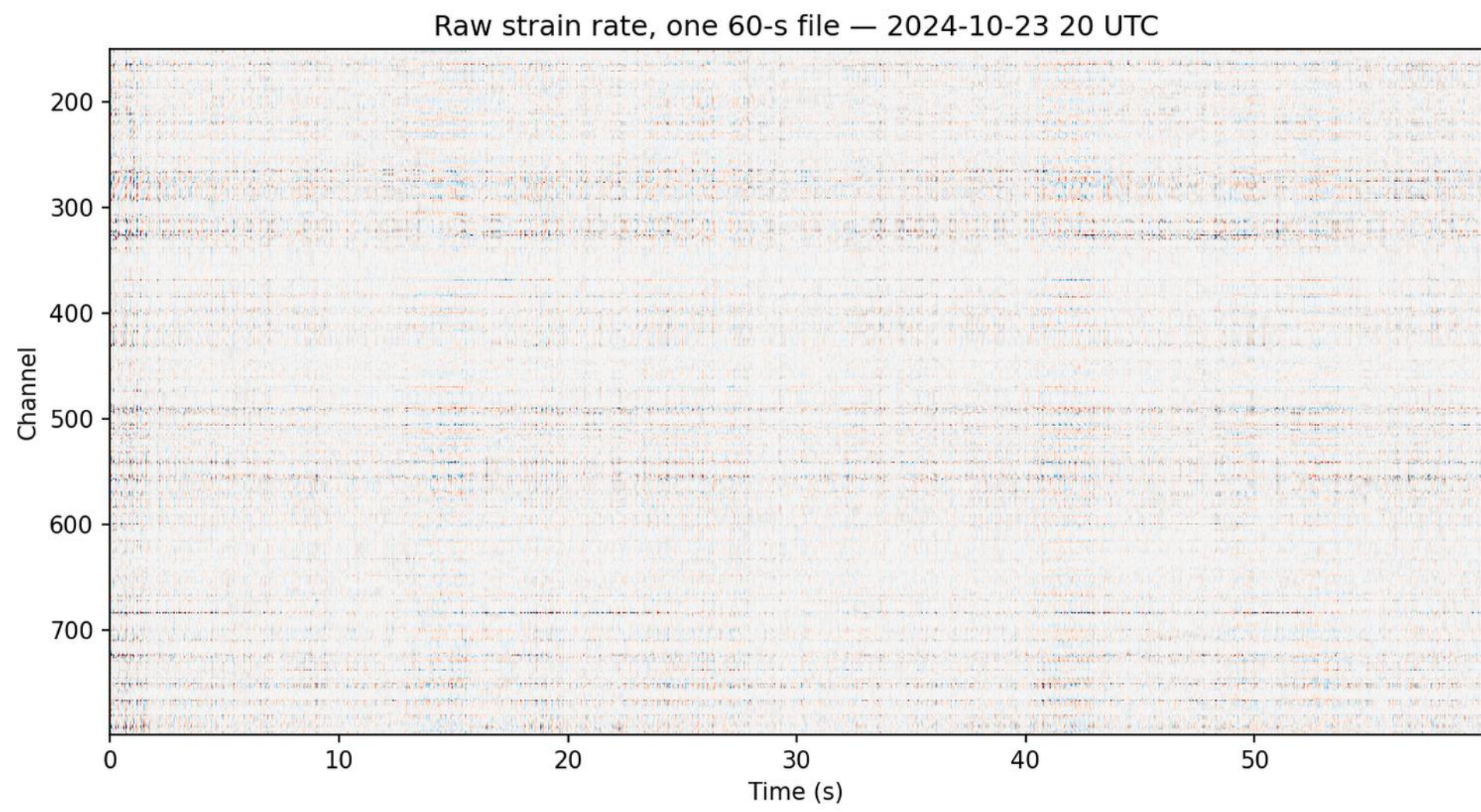
The preflight diagnostics were useful for deciding whether the run was worth attempting, but they should not be over-read as a successful Green's-function recovery. They characterize the raw wavefield, not the final stacked correlogram.



Preflight RMS per channel. The one-hour diagnostic showed a prominent high-RMS channel near ~310. The later CC job's automated first-files RMS pass flagged channels 705, 715, 721, and 731, so bad-channel behavior appears file/time dependent.



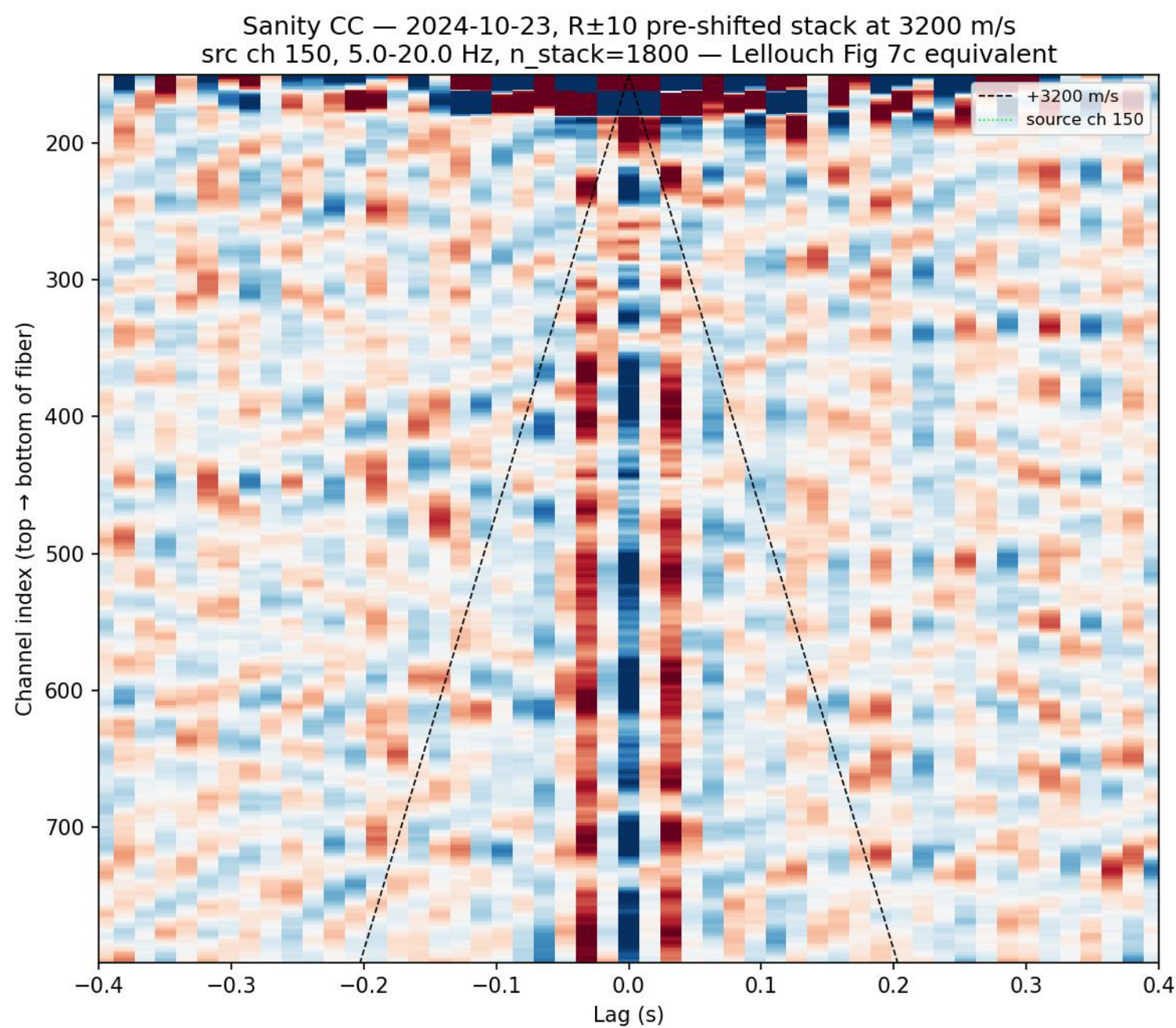
Preflight F-K spectrum. Energy exists in the 5-20 Hz band and is not purely surface-wave dominated, but the spectrum is broad enough that CC/post-processing is still the decisive test.



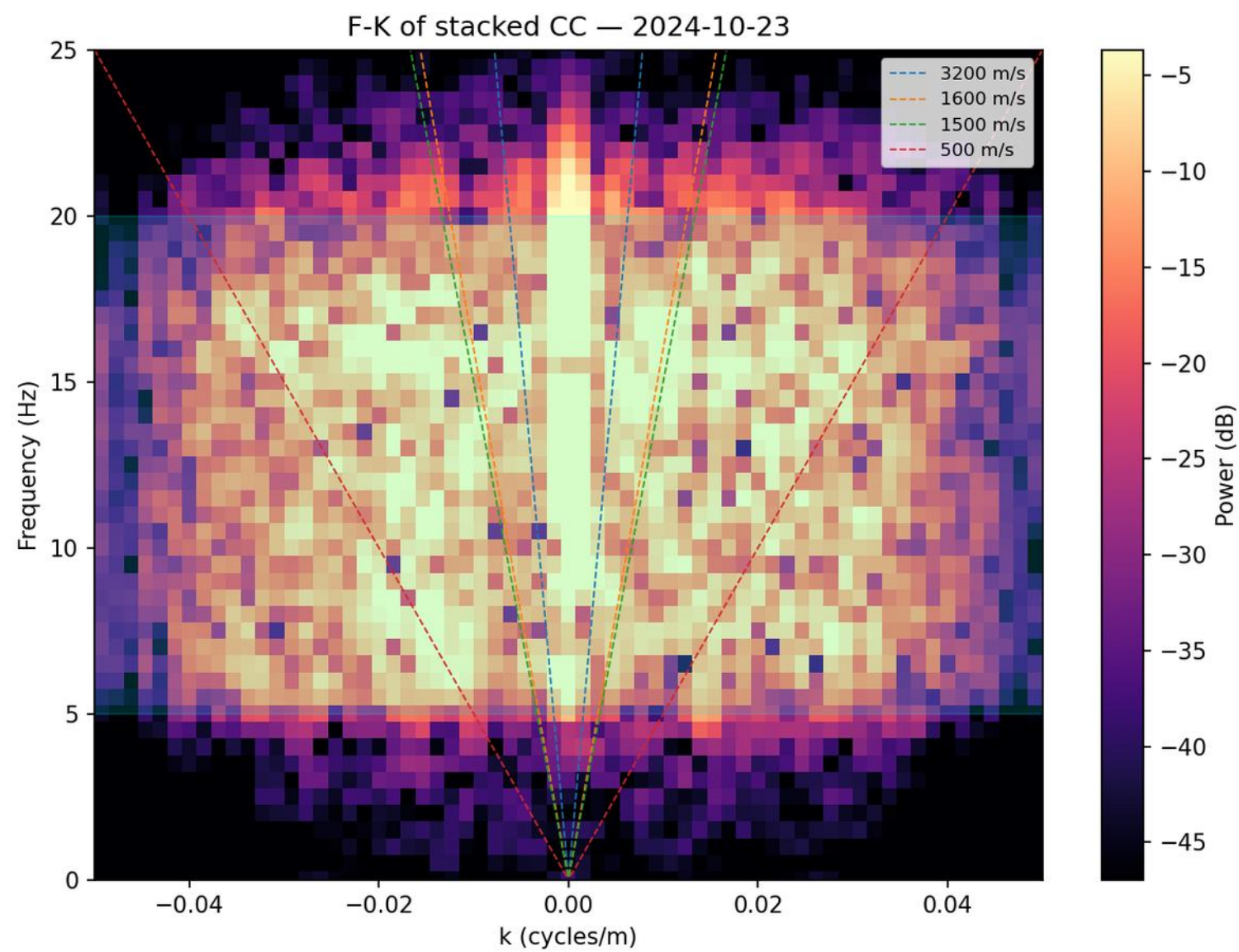
Raw time-distance panel for the same preflight hour. Useful for spotting coherent bands, bad traces, and common-mode structure before committing cluster time.

One-Day CC Result

The one-day CC completed cleanly. The key log facts were: 600 daytime continuous files selected, fs=62.5 Hz, ch 150-800, source ch 150, n_stack=1800, whitening off, and output saved to the sanity_v1 directory. The job warning from temporal normalization was handled with nan_to_num and was not a crash.



R±10 pre-shifted stacked CC for 2024-10-23. Dominant near-zero-lag vertical energy is visible; no clean coherent arrival follows the +/-3200 m/s reference moveout.



F-K of stacked CC. The result is broad/smeared, with strong near-zero-wavenumber/near-zero-lag structure rather than a clean 3200 m/s ridge.

Conclusion from this run

The one-day daytime-only run did not reproduce Lellouch Fig. 7c. The careful conclusion is not 'the 2024 data cannot work'; it is 'this particular daytime one-day source-channel-150 run did not recover the target body-wave moveout.'

Scientific Clarifications

Gauge Length

The corrected gauge-length model is $H(k_z) = i k_z \cdot \text{sinc}(k_z L / 2)$, where $k_z = 2\pi f / v_{\text{app}}$. In the 5-20 Hz band, with L about 16 m, the sinc term is near 0.98 for a 20 Hz, 3200 m/s body-wave-like arrival and near 0.93 for a 20 Hz, 1500 m/s guided/tube-like arrival. That is a small effect. It slightly suppresses slower modes more; it does not amplify tube waves.

F-K Filtering

F-K analysis maps energy by temporal frequency and spatial wavenumber along the vertical fiber. It can diagnose or isolate apparent velocities already present in the data. It cannot create a missing body-wave Green's function. It may later be useful for separating modes, but it should not be used to quietly redefine the strict reproduction test.

Surface-Wave Hydrology dv/v

The advisor's geometry concern is mostly right for standard ambient-noise surface-wave hydrology dv/v : a vertical borehole fiber does not sample lateral Rayleigh/Love paths the way a surface array or horizontal DAS line does. Hydrology-related monitoring could still be possible with body waves, local-event coda, repeating earthquakes, or guided/tube waves, but the sensitivity and interpretation would be different.

Code And Git Changes

Area	Change	Status
Multi-day/source-channel support	Added SANITY_DATES and SANITY_SOURCE_CH to the sanity CC pipeline; plot filenames now include channel/source tags to avoid collisions.	Pulled on Sherlock via GitHub update 7d8e9d9.
Gauge-length docs	Updated README language to stop claiming 16 m gauge length amplifies tube/casing modes in the 5-20 Hz CC band.	Committed/pulled in the same update.
All-hours support	Locally added SANITY_HOURS=all day night with default all; output filenames include hoursall/hoursday/hoursnight.	Local laptop change at report time; needs commit/push/pull before Sherlock can use it.
Sherlock runtime path	Sherlock needs /home/groups/ettore88/nberrios/safod_das_git/DAS-utilities/python because the new clone lacks DAS-utilities/python.	Manually restored on Sherlock from /tmp/run_sanity.sbatch.before_pull; should be committed intentionally later.

Sherlock State

- The new Sherlock clone is /home/groups/ettore88/nberrios/safod-das.
- The legacy repository /home/groups/ettore88/nberrios/safod_das_git still supplies DAS-utilities/python.
- A stash named 'preserve sherlock run_sanity edit' exists from protecting the local run_sanity.sbatch edit before pulling.
- Untracked sanity/plot_out/ and sanity/preflight_out/ directories exist on Sherlock; they are outputs and should not be added to Git.
- The one-day daytime CC output and plots are local evidence, but the all-hours code was not yet active on Sherlock at the time the week-long job was submitted.

Running Job

Job 24597004 was submitted as a week-long stack using SANITY_DATES=2024-10-22:2024-10-28, ch 150-800, source ch 150. Because the all-hours code had not yet been pulled to Sherlock, this job should be labeled as the week-long daytime-only stack.

```
# Check status/progress
cd /home/groups/ettore88/nberrios/safod-das/sanity
squeue -u nberrios
tail -20 logs/safod_sanity_24597004.err

# When CC finishes, plot the most recent npz produced by the current code
sbatch --export=ALL,STAGE=plot run_sanity.sbatch
```

Recommended Next Tests

1. Let job 24597004 finish. Treat it as the week-long daytime-only cultural-noise stack, not the Lellouch-faithful all-hours run.
2. Commit and push the local SANITY_HOURS change from the laptop, then pull it on Sherlock. Confirm the Sherlock run_sanity.sbatch still includes the legacy DAS-utilities path.
3. Run the all-hours week stack with SANITY_HOURS=all. This is the fairer Lellouch-style multi-day test.
4. Run day and night week stacks only as source-timing comparisons after the all-hours result exists.
5. Run source-channel sweeps for at least source channels 150, 180, 200, and 250 under the same all-hours setup.
6. Run the same code on the 2017 Lellouch-era data once its path/manifest is known. If 2017 reproduces and 2024 does not, the result becomes a strong 2017-vs-2024 source/fiber/wavefield comparison.

Exact Commands For The Next Clean All-Hours Run

Use these only after the SANITY_HOURS code has been committed, pushed, pulled on Sherlock, and the legacy DAS-utilities path has been preserved in the active run_sanity.sbatch.

```
cd /home/groups/ettore88/nberrios/safod-das/sanity

export SANITY_DATES=2024-10-22:2024-10-28
export SANITY_HOURS=all
export SANITY_CH_START=150
export SANITY_CH_END=800
export SANITY_SOURCE_CH=150

sbatch --export=ALL,STAGE=cc run_sanity.sbatch
```

Expected Output Naming

```
/oak/stanford/groups/ettore88/nberrios/sanity_v1/
sanity_cc_2024-10-22_to_2024-10-28_7d_hoursall_ch150-800_src150.npz

/home/groups/ettore88/nberrios/safod-das/sanity/plot_out/
cc_raw_2024-10-22_to_2024-10-28_7d_hoursall_ch150-800_src150.png
cc_stacked_2024-10-22_to_2024-10-28_7d_hoursall_ch150-800_src150.png
cc_fk_2024-10-22_to_2024-10-28_7d_hoursall_ch150-800_src150.png
```

Decision rule

Do not treat one failed daytime run as the final outcome. The stronger decision point is: all-hours multi-day + source-channel sweep + 2017 same-code test.